Feature Analysis for Classification of Physical Actions Using Surface EMG Data

Anish C. Turlapaty, *Member, IEEE*, and Balakrishna Gokaraju, *Member, IEEE*

***Abstract*— Based on recent health statistics, there are several thousand people with limb disability and gait disorders that require medical assistance. A robot assisted rehabilitation therapy can help them recover and return to a normal life. In this scenario, a successful methodology is to use an EMG signal based information to control the support robotics. For this mechanism to function properly, the EMG signal from the muscles has to be sensed and then the biological motor intention has to be decoded and finally the resulting information has to be communicated to the controller of the robot. An accurate understanding of the motor intention requires a pattern recognition based framework. Hence in this paper, we propose an improved classification framework through identification of the relevant features that drive the pattern recognition algorithm. Major contributions include a set of modified spectral moment based features and another relevant inter-channel correlation feature that contribute to an improved classification performance. Next, we conducted a sensitivity analysis of the classification algorithm to different EMG channels. Finally, the classifier accuracy with probabilistic neural networks based classifier is 92.7% and with an ensemble of subspace K-nearest neighbors algorithm it is 93.9% and is better than that of the other state of the art algorithms.**

***Index Terms*— PNN, feature combinations, classification, EMG, spectral moments.**

I. Introduction

A. Background

Physical disabilities have been a major problem in the modern world due to various reasons. For example, the aging brings problems such as gait disorders and limb impairments leading to a loss of quality of life [1]. Next occupational, trauma, and sports-based injuries and other kinds of severe accidents usually make people either completely or partially disabled. Another major cause of the disabilities in adults is a stroke to the motor cortex or other related regions. For example, an ischemic stroke can lead to motor disabilities, most commonly to the upper limbs [2]. In terms of the statistics, there are 300 amputees per year in the United Kingdom [3] and a total of 185, 000 amputees live in the United States [4]. Most of them need a prosthetic limb or partial limb support. Only a few treatment options are available for these individuals. For instance, a therapeutic rehabilitation can assist the partially disabled people for functional recovery in order to resume normal activities and thus improve the quality of life. In this context, wearable robots can assist in improving the effectiveness of rehabilitation [5]. There is growing evidence that a robot-assisted therapy provides an improvement of motor skills [6], [7]. The key reason for this improvement is due to the increased therapeutic repetitions and enhanced motivation for the patient due to the involvement of the virtual reality and video gaming [8]. The wearable robotics can also be used to ease the burden on human beings in various manual tasks. For example, the disabled can be supported by an externally powered prosthetic or an orthotic exoskeleton to restore the limb functionality to some extent [9], [10].

In most of these exoskeleton applications, a key goal is to build a human robot interface that can learn the intention of a user and adapt itself in order to provide accurate and timely assistance [11]. Note that in a healthy human, there is active communication between the central nervous system via motor neurons with the concerned muscle groups to produce the intended motion [12]. However, in an amputee or in a person with a stroke-induced disability, the bioelectrical signal may not reach the muscles or the muscles may not produce the required force. In these scenarios, an engineering solution is to sense the required myoelectric signal, assuming the signal has a causal relation to the intended motion, using an EMG sensor on the muscle surface and extract the relevant information. For example, one can estimate the joint force and decode the intended direction needed for the movement and relay it to the controller to generate the intended motion [13], [14]. The EMG signal basically refers to the electrical activity associated with the muscles when the muscle fibers are recruited via neuronal firings [15]. A practical exoskeleton product based on surface EMG, invented by an MIT based startup, Myomo, assists people with partial disabilities to perform their daily activities [16]. In this type of research, a major challenge is the design of suitable algorithms and the hardware to cope up with the complexity of human systems and further adapt in an uncertain environment [17]. In recent years, there is growing evidence that the pattern recognition (PR) algorithms can be used to learn the motor intention from the surface EMG signals [18]. In the next subsection, we present a survey of the most recent algorithms for PR based control of exoskeletons.

B. Literature Review

This technology usually has (1) a software component in which an algorithm does relevant feature extraction and

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selection and categorization of movements using a statistical discriminator and (2) a hardware component to interface with an exoskeleton via a controller. In the literature, as discussed below, we find a great deal of emphasis on the algorithm part specifically on the feature extraction. In [12], an online method was developed that has a PR stage and a control mechanism. The classification was performed for 8 hand gestures using the RMS value as a feature and a linear SVM achieving 95% accuracy. Next, an orthotic arm guided by an EMG based control with up to four degrees of freedom was demonstrated for performing basic manoeuvres in a 3D environment such as simple arm movements and object pick and place. In [19], an ANN-based control was developed that has a sensitivity to the multiple levels of movements of flexors and extensors to directly control the orthotic arm. In [14], a PR algorithm was developed to determine the torque level from a human wrist and use it for real time control of an exoskeleton prototype. It was shown that a set of four EMG channels is sufficient for the classification of different wrist actions at different torque levels. In [3], a classification experiment was done, in which three healthy subjects and three amputees participated. The healthy subjects were asked to perform 15 different finger movements and the amputees were asked to imagine 12 different movements. Using the frequency domain features and the SVM, the movements were classified with an accuracy greater than 90%. In [20], the intrinsic mode functions were used for feature extraction toward the classification of six categories of the handgrips. In another work, the cardinality of the EMG signal was proposed as a highly relevant feature for identifying the motor intent [21]. Next, in [4], a classification of six handgrips and finger positions was analysed against the force levels. Nine amputees participated in the study and the key innovation is the introduction of the moment ratios as discriminative features for a successful classification with accuracy over 90% despite discrete variations in the force levels. Next, in [22] an analysis was performed of the relevant features and the selection of corresponding classifiers for control of the exoskeletons for partially disabled hands. They addressed the classification of EMG signals corresponding to four hand positions for 20 subjects of which 16 are non-amputees and 4 are partial hand amputees. The efficiency of the time domain features was compared with the autoregressive features and the frequency domain features. Next in [23], the local binary pattern based features were used for the classification of the physical action EMG data [24]. A major drawback in [23], is that it is not clear whether the analysis is for a binary classification i.e, the normal vs. the aggressive or for classification of all the 20 categories of the data. In the current paper, as described in the following sections, we address the above mentioned multi-category classification problem.

C. Contributions

In our paper, we address the classification of M-categories of physical actions based on multi-channel EMG data. We propose an improved feature set consisting of selected feature subsets from different feature modalities such as the time domain (TD) statistics, the inter-channel TD statistics, the spectral moment ratios and products, the spectral band powers and the local binary pattern based statistics. In the feature extraction apart from other well known features, we have modified the spectral moment features to improve the classification performance. Moreover we have identified an inter-channel correlation feature that also contributes to an improved classification. Next, we present the key EMG channels that significantly contribute toward the performance improvement. Finally, we also demonstrate that an Ensemble of subspace KNN, a single KNN and a probabilistic neural network results in an accuracy $\geq$ 92.7% and an area under the ROC curve $\geq$ 0.974.

II. Methodology

We consider a dataset $\mathbf{X}$ with $P(= S \times C \times R)$ observation arrays obtained from S subjects. These observations consist of C categories with R trials each ie., each of the S subjects has performed each activity R times as represented in fig. 1. Finally, each of the p-th observation array X_p has M channels (distinct EMG electrode contact locations) and each of the m-th channel $\mathbf{x}_p^m$ has N values (samples).

$$\begin{aligned}
\mathbf{X} &= \left\{ X_1, X_2, \cdots, X_P \right\} \\
X_p &= \left[\mathbf{x}_p^1, \mathbf{x}_p^2, \cdots, \mathbf{x}_p^M \right], \quad p = 1, \cdots, P \\
\mathbf{x}_p^m &= \left[x_p^m(1), x_p^m(2), \cdots, x_p^m(N) \right]^T, \quad m = 1, \cdots, M
\end{aligned} \tag{1}$$

The first step is segmentation of the EMG signal in each channel with a non-overlapping and a sliding window length L, which is derived based on the sampling rate of the concerned EMG data. The standard window length (in time) is 200ms from literature, see [14] and [3]. Next, the l-th element of the w-th segment of the m-th channel of the p-th observation is denoted as

$$s_{\mathbf{j}}(l) = x_{\mathbf{j}}((w-1) * L + l) \tag{2}$$

here, $\mathbf{j}$ represents the index triplet $\mathbf{j} = (m, w, p)$, where $w = 1, \cdots, N_w, l = 1, \cdots, L$ and $N_w = \frac{N}{L}$. Note that each pattern can consist of N_s segments and features from each segment are combined to build the complete feature vector corresponding to a pattern.

Now we give an overview of the proposed pattern recognition framework as given in fig. 2. The EMG signals are segmented and the features from various modalities are extracted as described in the subsequent sections. The combined feature set is given as an input to a probabilistic neural network for the classification. Next the relevant features are selected using a forward feature selection algorithm. The performance is analyzed using a 10-fold cross validation. Finally, the classification performance is compared with those of several different classifiers. Next, we give the details of the feature extraction algorithms.

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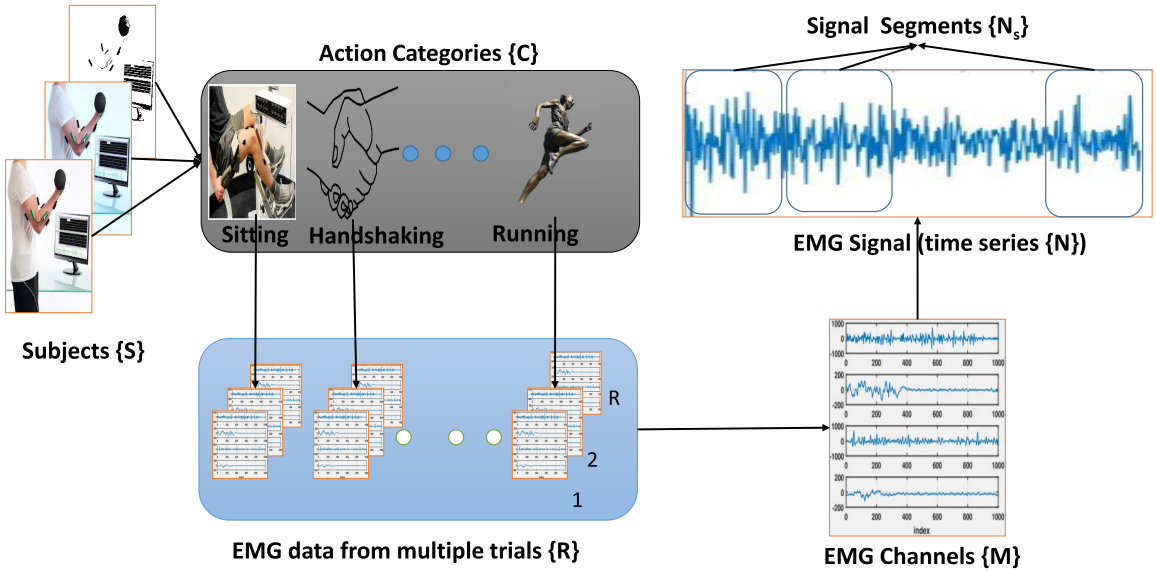


Fig. 1. Physical action data: setup for classification.

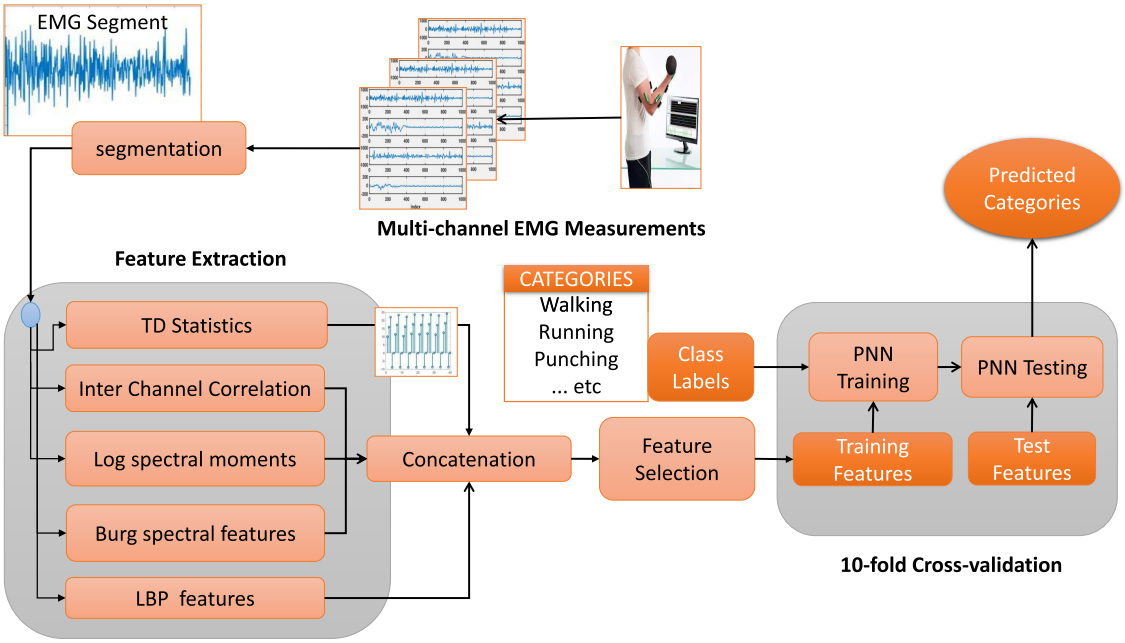


Fig. 2. Physical action data classification scheme using PNN.

A. Feature Extraction

1) Time Domain Statistics (TDS): The first subset of features is computed from the sample statistics for each segment within a pattern [25]. The mean value for the **j**-th segment (see eq. (2)) is defined as

$$f_t(\mathbf{j}) = \frac{1}{N_w}\sum_{l=1}^{N_w} s_{\mathbf{j}}(l) \tag{3}$$

Similarly, the variance, skewness and kurtosis are computed. Finally, for a p-th pattern, the 4 features from each segment and M channels are grouped into the TDS feature vector $\mathbf{f}_{TDS}(p)$.

2) Inter-Channel Statistics (ICS): This subset of features is based on maximum cross-correlation [26] among the corresponding w-th segments of the two channels i and j of the p-th signal and is defined as

$$\rho_{w,p}^{i,j} = max\left(E\{s_{i,w,p}(l)s_{j,w,p}(l)\}\right) \tag{4}$$

Note that the cross-correlation function need not have a maxima at the zero-lag. Here the values of i and j are chosen from the set C_1 of $\binom{M}{2}$ ordered pairs defined below.

$$C_1 = \{(i,j) : i = 1, 2, \cdots, M-1;\ j = 2, 3, \cdots, M \text{ and } i \neq j\} \tag{5}$$

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The feature vector corresponding to the M channels is

$$\mathbf{f}_{ICS}(p) = \{\rho_{w,p}^{i,j};\, (i,\, j) \in C_1\} \tag{6}$$

3) Log Moments of Fourier Spectra (LMF): The logarithms of moments and their ratios from the frequency domain are computed for the EMG segments based on [3]. Consider the L-point Fourier transform of a segment in the m-th channel of the p-th EMG signal

$$S_{\mathbf{j}}(k) = \sum_{l=1}^{L} s_{\mathbf{j}}(l) \exp\left(-\frac{j2\pi lk}{L}\right), \quad k = 1, \cdots, L \tag{7}$$

Now the squared magnitude spectrum is defined as

$$\psi_{\mathbf{j}}(k) = |S_{\mathbf{j}}(k)|^2 \tag{8}$$

The i-th frequency domain moment from the spectrum is defined as [4]

$$g_{\mathbf{j}}(i) = \sqrt{\sum_{k=1}^{L} k^i \psi_{\mathbf{j}}(k)} \tag{9}$$

where $i \in \{0, \cdots, 6\}$. Now, the moment features are defined as

$$\begin{aligned}
f_{\mathbf{j}}(1) &= \ln g_{\mathbf{j}}(0) \\
f_{\mathbf{j}}(2) &= \ln g_{\mathbf{j}}(2) \\
f_{\mathbf{j}}(3) &= \ln g_{\mathbf{j}}(4) \\
f_{\mathbf{j}}(4) &= \ln g_{\mathbf{j}}(0) - \frac{1}{2}\ln\big(g_{\mathbf{j}}(0) - g_{\mathbf{j}}(2)\big) \\
&\quad -\frac{1}{2}\ln\big(g_{\mathbf{j}}(0) - g_{\mathbf{j}}(4)\big) \\
f_{\mathbf{j}}(5) &= \ln g_{\mathbf{j}}(2) - \frac{1}{2}\ln\big(g_{\mathbf{j}}(0) g_{\mathbf{j}}(4)\big) \\
f_{\mathbf{j}}(6) &= \ln g_{\mathbf{j}}(0) - \frac{1}{4}\ln\big(g_{\mathbf{j}}(1) g_{\mathbf{j}}(3)\big) \\
f_{\mathbf{j}}(7) &= \ln g_{\mathbf{j}}(0) - \frac{1}{4}\ln\big(g_{\mathbf{j}}(2) g_{\mathbf{j}}(6)\big)
\end{aligned} \tag{10}$$

The pair wise features based on moment products are

$$f_{\mathbf{j}}(n) = \frac{1}{2}\ln g_{\mathbf{j}}(i) g_{\mathbf{j}}(j) \text{ with } n = 8, \cdots, 17 \tag{11}$$

where the values of i and j are chosen from the set C_2 of 10 ordered pairs defined below.

$$C_2 = \{(i,\, j) : i = 1, 2, 3, 4;\, j = 2, 3, 4, 5 \text{ and } i \neq j\} \tag{12}$$

Note that the features $f_{\mathbf{j}}(n)$ for $n = 6, \cdots, 17$ are newly proposed. For the M channels, the features defined in eqs. 10 and 11 are grouped into the feature vector $\mathbf{f}_{LMF}(p)$.

4) Spectral Band Powers (SBP) : The spectral features were previously proposed for the EMG pattern recognition in [27]. In this work, the spectral band power features are extracted as follows. For each channel of the p-th pattern, assuming a model order ν, the auto-regression model coefficients $\{a_i\}$ are computed using the Burg's method described in [28].

$$H(\omega_k) = \frac{1}{A(\omega_k)} \tag{13}$$

here $A(\omega)$ is derived from the z-transform $A(z) = \sum_{i=0}^{\nu} a_i z^i$. Next the power spectral density estimate is given by

$$\Psi(\omega_k) = |H(\omega_k)|^2 \sigma_{burg}^2 \tag{14}$$

here σ_{burg}^2 is the error variance computed in the Burg's method. Finally the SBP features are evaluated by dividing the spectrum into N_b bands and computing the respective powers in those bands. For the bth band, the spectral index is defined as $k : K_{b-1} + 1, \cdots, K_b$ where K_b is the length the bth band. The power within the band is

$$\eta_b^m = \sum_{k \in b^{th} band} \Psi(\omega_k), \quad b = 1, \cdots, N_b \tag{15}$$

and the feature vector $\mathbf{f}_{SBP}(p)$ is composed of these band powers for all b and m.

5) Local Binary Patterns: For the w-th segment of the m-th channel in the p-th pattern, the local binary patterns (LBP) can be computed as follows

$$s_{lbp}(i) = \sum_j \Phi_j^{b(i)} \tag{16}$$

here, Φ_j are the elements of the basis vector used in computing the LBPs for the i-th interval. Next the exponents $b(i)$ are given by

$$b_i = u\big(\{g_i - g_c\}\big) \tag{17}$$

here $u(\cdot)$ is the step function and g_i are the values of the signal over a sliding window of length N_{LBP} and g_c is the mean of a subset of the same vector, for details refer to [29]. Finally, the features are computed by counting the number of LBP values above and below a certain threshold τ_{lbp}.

$$\begin{aligned}
f_{lbp}^m(1) &= \#\{s_{lbp} \leq \tau_{lbp}\} \\
f_{lbp}^m(2) &= \#\{s_{lbp} > \tau_{lbp}\}
\end{aligned} \tag{18}$$

here $\#\{A\}$ represents the cardinality of a set A. Finally, the feature vector $\mathbf{f}_{LBP}(p)$ is composed of these LBP features for all M channels.

B. Learning Using Classifiers

In this work, we implemented the classification using several existing classifiers with a focus on the probabilistic neural networks. The application of PNN for EMG signal classification is discussed in [30]. The learning methodology for the multi-class problem using the PNN is based on learning the multi-modal multi-variate PDF of the data using the Parzen's method. Ideally, each category has a corresponding mode in the multi-modal pdf. For the test data, the posterior probabilities that a pattern belongs to each of these modes are computed. Next, based on the maximum among these posterior probabilities, the corresponding class is assigned to the test pattern. The key feature of the PNNs is that there is a separate neuron for each training pattern and hence the size of the network depends on the size of the training data [31].

The other classifiers used for bench-marking the performance are (1) a single K-Nearest Neighbor classifier (fKNN) [32], (2) an ensemble of subspace KNNs (sKNN) [32], (3) a SVM with a cubic polynomial kernel (cSVM) [33], (4)a SVM with a Gaussian Kernel (gSVM) [33], (5) the Bagged Trees (BTR) [34] and (6) the Linear Discriminant Analysis (LDA) [35]. The performance of the classifiers is

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TABLE I
PHYSICAL ACTION CATEGORIES AND CLASS LABELS

Label	Normal	Label	Aggressive
1	Bowing	11	Elbowing
2	Clapping	12	Frontkicking
3	Handshaking	13	Hammering
4	Hugging	14	Headering
5	Jumping	15	Kneeing
6	Running	16	Pulling
7	Seating	17	Punching
8	Standing	18	Pushing
9	Walking	19	Side-kicking
10	Waving	20	Slapping

TABLE II
CARDINALITIES OF EACH FEATURE SUBSET THROUGH COMBINATION OF THE 8 CHANNELS

Feature Subset	TDS	ICS	LMF	SBP	LBP
No. of features	32	12	136	80	16

evaluated using the classification accuracy (α), Kappa coefficient (κ), $F1$ measure and multi-class area under ROC curve (γ) [36].

III. IMPLEMENTATION AND RESULTS

A. Physical Action Dataset

The data set of interest is taken from the UCI Machine learning repository [24]. The dataset consists of the EMG signals recorded using the Delsys EMG electrodes on $S = 4$ subjects while they performed $C = 20$ different physical activities of which 10 were aggressive and 10 were normal activities as listed in table I. Each subject repeats the physical action $R = 15$ times. There were 8 EMG electrodes placed on each subject, first 4 on the biceps and triceps muscle groups of the upper limbs and next 4 on the thighs and hamstring muscle groups of the lower limbs. Finally, each channel consists of approximately $10,000$ values. Hence, in this study the total sample size becomes $P = 4 \times 20 \times 15 = 1200$.

B. Application of Classification

The features from $M = 8$ channels for each modality are computed as follows. Based on $R = 15$ and total signal length 10000, the EMG signal length per trial is $N = 666$. The number of segments is $N_s = 1$, and the width $N_w = N$, Next, the feature subsets from the different feature extraction modalities mentioned in section (II-A) are computed and combined into a full feature vector.

$$\mathbf{f}_{all}(p) = \left[\mathbf{f}_{TDS}(p),\ \mathbf{f}_{ICS}(p),\ \mathbf{f}_{LMF}(p), \mathbf{f}_{SBP}(p),\ \mathbf{f}_{LBP}(p)\right] \tag{19}$$

As presented in the table II, the cardinality of each subset (except ICS) is $8 \times c_{mod}$, where c_{mod} is the cardinality of the feature subset per modality per channel. For example, for time domain features, the cardinality is $8 \times 4 = 32$.

Now, based on the pairs given in table III, the cross-correlation features (6 each) are computed separately for the subset of the channels from the upper limbs and those from the lower limbs. The LMF cardinality is based on the fact that each channel has 17 LMF features amounting to a total of 136 features. For the spectral band powers, the number of bands is chosen to be $N_b = 10$ leading to a total of 80. Finally, the 2 LBP features are computed per channel counting to a total of 16.

TABLE III
CHANNEL PAIRS CONSIDERED FOR COMPUTING THE CROSS-CORRELATION FEATURES

Index	Upper Limbs	Index	Lower Limbs
1	(3,4)	7	(7,8)
2	(2,4)	8	(6,8)
3	(2,3)	9	(6,7)
4	(1,4)	10	(5,8)
5	(1,3)	11	(4,7)
6	(1,2)	12	(5,6)

TABLE IV
SELECTED FEATURES (INDEX) FROM EACH MODALITY TAKEN FROM THE 8 CHANNELS THAT HAVE SIGNIFICANT CONTRIBUTION TO THE CLASSIFICATION PERFORMANCE

Subsets	TDS	ICS	LMF	SBP	LBP
Chan					
1	1	6	4, 10, 11, 14, 16	-	-
2	7	3, 6	24, 29, 30, 34	11, 13	-
3	9,10, 11	3	41, 42, 48	23,26,30	5
4	-	-	-	31	-
5	17	12	78	-	9
6	23	12	-	-	-
7	25	-	107	66	13
8	29	-	126	-	-

C. Analysis

Initially, the generated feature vector consisting of 276 features is used as an input to the PNN classifier. In the cross-validation strategy, the sequence of the feature vectors and the output labels is shuffled to avoid any learning bias in the classification. Next, the algorithm performance is evaluated by averaging the 10 fold cross-validation based classification metrics across $M_c = 100$ Monte-Carlo runs. The average classification performance when all the feature subsets are used is $\alpha = 93\%$ and $\kappa = 0.925$.

1) Sequential Forward Feature Selection: Now we apply the forward feature selection process to the feature set $\mathbf{f}_{all}$ as follows. The process begins by selecting a feature that has the best accuracy among the individual features. This feature is placed in $\mathbf{f}_{sel}$, the selected feature subset. In the next step, a second feature that improves the performance is selected and added to the same subset $\mathbf{f}_{sel}$ and this process is repeated until the classification performance converges to the global maximum. The list of selected features is presented below. The feature index in table IV is based on the cardinality of the subsets given in table II.

Among the time domain statistics, the mean value of the EMG signal is the key feature selected from the channels $1, 3, 5, 7, 8$. Next the variance statistic from the channel 3 is selected. Followed by the skewness from the channels $2, 3$ and 6. Finally the kurtosis is not selected and hence irrelevant for improving the classification accuracy.

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TABLE V

CHANNEL PAIRS THAT HAVE SIGNIFICANT CROSS-CORRELATION AND CONTRIBUTE TO PHYSICAL ACTION CLASSIFICATION

Upper Limbs	Lower Limbs
(1,2); (2,3)	(5,6)

TABLE VI

SETUP INFORMATION FOR EACH OF THE CLASSIFIERS USED IN BENCH-MARKING

Classifier Name	Model Setup
PNN	Spread = 1
sKNN	Splits = 20, Learners = 60, learning rate = 0.1 , subspace dimension = 19
fKNN	Distance measure = Cosine, distance weights = equal, $K = 3$
BTR	No of splits = 1199, no. of learners = 30, learning rate = 0.1
cSVM	Polynomial kernel, Order = 3, Scale = 1, Box constraint = 1
gSVM	Gaussian Kernel, Scale = 6.1
LDA	Full covariance

Next, in terms of the inter channel statistics, the correlation between the EMG signals of the channels $(1, 2)$ (i.e., the right biceps and triceps) and $(2, 3)$ (i.e., the right triceps and left biceps), and again the correlation between the channels $(5, 6)$ (i.e., the right thigh and hamstring) have been found to be contributing to the classification (see table V). Next, among the FFT based moment ratios/products (LMF), the relevant features in each channel are specified in the table IV. Clearly the moment ratio defined as $f_{\mathbf{j}}(7)$ given in eq. 10 is the most relevant feature among the LMF features. Next in the channels $1, 2$ the moment ratios defined in eq. 11 are found to be of relevance. Next the moment product features $f_{\mathbf{j}}(10)$ in the eq. 11 from the channel 5 and $f_{\mathbf{j}}(5)$ from the channel 7 also contribute to the classification. Finally, among the LMF features, those from the channels 4 and 6 are found to be irrelevant. From the SBP feature selection we note that only the average powers from the bands $2, 3, 4$ and 7 as mentioned in table IV are found to be relevant. Finally, in the case of the LBP features, the features $f_{lbp}(1)$ from the channels $3, 5$ and 7 are contributing to the classification.

2) Channel Relevance Analysis: The relevance of the channels toward classification is measured in terms of α and κ. In this approach, we perform the classification of the EMG data, omitting one channel at a time from the selected features as given in table IV, to compute the α and κ. From $M_c = 100$ runs of each of the 8 cases, we obtain the average classifier performance in a $10-$fold cross validation framework and present it in fig. 3. From this figure, it is clear that the channels $1, 2$ and 3 i.e., the features of the EMG data from the electrodes placed on the right biceps, right triceps and left biceps have the most relevance. Next features corresponding to the channels 7 i.e., left thigh can be ranked second. Finally, the channels $4, 5, 6$ and 8 (left triceps, right thigh, right hamstring and left hamstring) fall third in terms of relevance toward classification.

3) Comparison Across Classifiers: The performance of the PNN classifier is compared to that of each of the classifiers

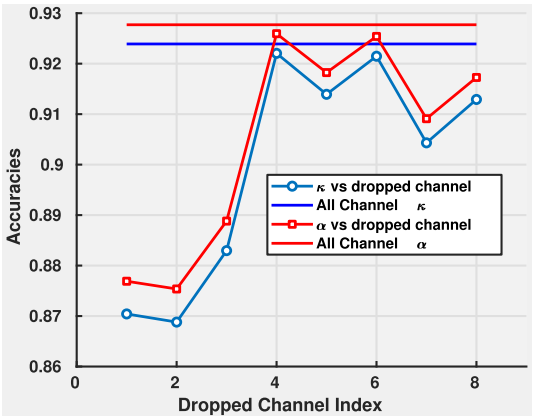
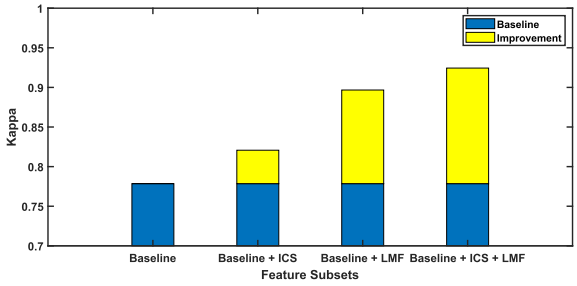
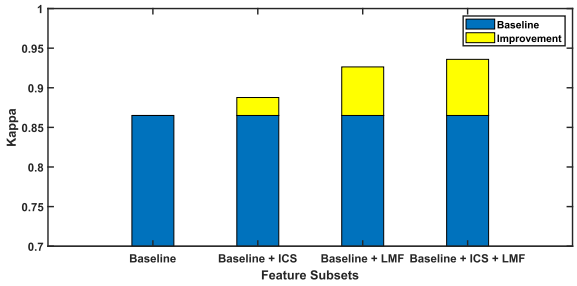


Fig. 3. Sensitivity of classification performance to EMG channels.



(a) PNN



(b) sKNN

Fig. 4. The performance improvement in κ due to inclusion of the proposed and selected features.

TABLE VII

CLASSIFICATION PERFORMANCE COMPARISON AMONG MULTIPLE CLASSIFIERS WITH THE SELECTED SUBSET OF FEATURES

Classifier Name	α	κ	$F1$	γ
PNN	92.78	.924	.929	.975
sKNN	93.91	.936	.939	.975
fKNN	92.96	.925	.93	.973
cSVM	92.22	.918	.923	.97
gSVM	91.26	.90	.913	.967
BTR	89.1	.885	.891	.958
LDA	71.09	.695	.717	.915

mentioned in section II-B. For each of these classifiers the setup including a choice of hyper-parameters is mentioned in the table VI. For bench-marking the performance of the classifiers, we have compared the classification metrics across seven different classifiers as shown in table VII. Moreover, the same set of selected features from the SFS algorithm

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TABLE VIII
CONFUSION MATRIX FROM PNN WITH 10-FOLD CROSS VALIDATION

Predicted	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
True																				
1	59		1																	
2		60																		
3		1	58							1										
4		2		58																
5					55	5														
6					1	59														
7							59	1												
8							3	57												
9									60											
10	1									59										
11											49	1	2	1	3		2			2
12												52	2						4	2
13												1	56				3			
14												1		53						7
15											1				55	2		2		
16											1					56	1	1		1
17													1				59			
18						1									2	4	1	53		
19											1	16		1					41	1
20												1		2						57

TABLE IX
CONFUSION MATRIX FROM ENSEMBLE OF SUBSPACE KNNS WITH 10-FOLD CROSS VALIDATION

Predicted	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
True																				
1	59		1																	
2		59		1																
3			59							1										
4		1		59																
5					57	3														
6					3	57														
7							59	1												
8								60												
9									60											
10										60										
11											52			1	4		1	1		1
12												51							9	
13													56				4			
14												1		56						4
15											2				56			2		
16											1					58		1		1
17													4				56			
18						1					1				1	4		53		1
19												9							51	
20														3			1		1	55

based on the PNN based approach are used in each of the other classifiers. Next we present an summary of different performance metrics

- *accuracy α and Kappa coefficient κ:* The sKNN classifier has the highest $\alpha = 93.9\%$ and $\kappa = 0.936$ followed by fKNN and PNN classifiers with $\alpha \geq 92.7\%$ and $\kappa = 0.924$. Performance of both the SVM algorithms is slightly lower by 2%. Finally, the LDA algorithm resulted in the lowest performance with $\alpha = 71\%$ and $\kappa = 0.695$.
- *F1-measure:* Again, the sKNN resulted in a maximum F1 measure of 0.939 with fKNN and PNN resulting in 0.924 (lower by 1%), followed by the SVM and again the LDA has the lowest F1 measure of 0.717.
- *Area under ROC curve γ :* In terms of γ , PNN and sKNN have similar performance $\gamma = 0.975$, followed by fKNN and the SVM algorithms and again the LDA has the lowest area $\gamma = 0.915$.

4) Comparisons Across Classes: The across class performance for PNN is given in the confusion matrix (CM) in table VIII. Recall that the physical actions have two broad categories, the normal and the aggressive. From the confusion matrix in table VIII, the first quadrant of the matrix has a significantly stronger diagonal compared to the fourth quadrant which suggests that the PNN classifier is highly successful in learning the normal actions compared to the aggressive actions. The second and the third quadrants of the CM are nearly all blank (i.e. zeros) suggesting that there is very little confusion between the normal and the aggressive activities. In the fourth quadrant, the section of the confusion matrix corresponding to the aggressive activities, shows that there is slightly more number of miss-classifications. For example, 16 patterns from the class 19 get classified as the class 12 which means there is a considerable confusion between the front kicking and the side kicking actions. Next, the confusion

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matrix resulting from the implementation of the classification using an ensemble of subspace KNN is presented in table IX. A similar analysis also applies to this and other classifiers. For example, as shown in the table IX, there is a confusion between the front kicking and the side kicking activities.

5) Feature Contributions: The selected features included a few features from each subset along with the newly proposed spectral moment ratio and products and the inter channel correlation features. In the fig. 4, the term *Baseline* refers to the subset of 21 selected features that are previously used in existing literature. Next, *ICS* refers to the selected inter-channel correlation features (see table V) and *LMF* refers to the proposed spectral moment ratios and products (see tables IV). As shown in fig 4a, for the PNN classifier, the inclusion of the *ICS* features improves the Kappa performance by 4.2%, the *LMF* by 11.8% and the union of the proposed subsets is 14.59%. Similarly for the sKNN classifier (see fig 4b), the improvements due to the ICS, LMF and the union of the subsets are 2.2%, 6.1% and 7.4% respectively.

IV. Conclusion

In this study, we have implemented a multi-category classification framework based on the probabilistic neural networks to categorize the physical actions using the features derived from eight channels of the surface EMG data. A set of 276 features were extracted from various modalities including the statistical features in time domain, the inter channel cross correlation features, logarithms of moment ratios of the Fourier spectra, the mean band powers of power spectral density estimates based on the Burg algorithm, and the features based on the local binary patterns.

Using the sequential forward selection algorithm a subset of 37 features from above 276 features were found to be relevant for the physical action classification. The selected features included a few features from each subset and also a set of the modified spectral moment products/ratios and the inter channel correlation features. Based on the selected features, the average classification performance metrics for the sKNN, PNN and fKNN are similar. Next the classification performance is slightly reduced for kernel based SVMs and further degrades for the Bagged Trees and the LDA algorithm. In terms of channel relevance, the EMG data from the upper limbs has greater significance in physical action classification. The future plans for this research have two important directions. The first is to explore the deep learning methods to improve the classification accuracy. A second direction is to conduct actual experiments to acquire the EMG measurements for controlling an orthotic exoskeleton for an upper limb. Finally, the overall goal is to find an optimal combination of learning algorithms and control strategies for upper limb exoskeletons.

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